

KESHAV BHANDARI

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<https://www.linkedin.com/in/bhandarikeshav/>

<https://keshavbhandari.github.io>

SUMMARY

I am a PhD student at the Centre for Digital Music (C4DM) at Queen Mary University of London. My research revolves around neuro-symbolic approaches for generating music with a focus on controllability and musical structure.

EDUCATION

Queen Mary University of London, EECS

London, UK

Ph.D. in Artificial Intelligence

2023 - 2027 (*expected*)

- Advisors: Simon Colton; Geraint Wiggins
- Research area: Neuro Symbolic Automated Music Composition

Northwestern University, McCormick School of Engineering

Evanston, USA

Master of Science in Artificial Intelligence (GPA: 3.96/4.0)

2021 - 2022

Purdue University, Krannert School of Management

Indiana, USA

Bachelor of Science in Business Management & Marketing

2012 - 2016

- Dean's List, 2013

PUBLICATIONS

1. **Bhandari, K.**, Bizzarri, M., Wiggins, G. A., & Colton, S. (2026). Change is Key: A Generative Framework for Controllable Musical Modulations. Manuscript under review, JNMR.
2. **Bhandari, K.**, Chang, S., Roy, A., Ronchini, F., Benetos, E., Herremans, D., & Colton, S. (2026). Text2Score: Generating Sheet Music From Textual Prompts. Manuscript under review, AAAI.
3. **Bhandari, K.**, Roy, A., Herremans, D., & Colton, S. Emerging AI Technologies for Music: Towards Controllable, Collaborative, and Creative Systems. *Proceedings of Machine Learning Research (PMLR)*, vol. 303, pp. 1–5, 1st Workshop on Emerging AI Technologies for Music (EAIM) at AAAI, 2026.
4. **Bhandari, K.**, Chang, S., Lu, T., Enus, F. R., Herremans, D., & Colton, S. ImprovNet - Generating Controllable Musical Improvisations with Iterative Corruption Refinement. *International Joint Conference on Neural Networks (IJCNN)*, 2025.
5. **Bhandari, K.**, Wiggins, G. A., & Colton, S. Yin-Yang: Developing Motifs With Long-Term Structure And Controllability. *International Conference on Artificial Intelligence in Music, Sound, Art and Design (EvoMUSART)*, 2025. (**Best Paper Award**)
6. **Bhandari, K.**, Roy, A., Wang, K., Puri, G., Colton, S., & Herremans, D. Text2midi: Generating Symbolic Music from Captions. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
7. Colton, S., Bradshaw, L., Banar, B., & **Bhandari, K.** Automatic Generation of Expressive Piano Miniatures. *International Conference on Computational Creativity (ICCC)*, 2024.
8. **Bhandari, K.**, & Colton, S. Motifs, Phrases, and Beyond: The Modelling of Structure in Symbolic Music Generation. *In International Conference on Artificial Intelligence in Music, Sound, Art and Design (EvoMUSART)*, 2024.
9. O'Reilly, P., Bugler, A., **Bhandari, K.**, Morrison, M., & Pardo, B. VoiceBlock: Privacy through Real-Time Adversarial Attacks with Audio-to-Audio Models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

PROFESSIONAL EXPERIENCE	Singapore University of Technology and Design Singapore Aug 2025 - Nov 2025	
	• Research Intern, AMAAI Lab (Advisor: Prof. Dorien Herremans) – Developed ImprovNet 2, a framework for genre-controllable musical improvisation using LLaDA (discrete diffusion models). – Built Text-to-Sheet-Music, a hierarchical model orchestrated by LLMs in a two-step decoding process, enabling text-conditioned music generation without explicit text–music annotations during training.	
	Boston Consulting Group Los Angeles, CA, USA Jan 2023 - Sep 2023	
	• Data Scientist – Vodafone Italy: Developed AI-based tools using Llama and Stable Diffusion to enhance digital marketing efficiency. – Staples: Designed a pattern matching algorithm to improve control group selection for precise reporting and strategic insights. – Walgreens: Built model drift scripts to rectify deviations in price optimization algorithm in a MLOps framework.	
	Epsilon Bangalore, Karnataka, India Sep 2017 - Feb 2021	
	• Senior Data Scientist – Built deep learning based personalized recommendation engines using Transformers, boosting client hit rates by up to 5%. – Deployed end-to-end ML & DL solutions as a product on cloud platforms, achieving lift rates between 3X to 9X. – Developed Gredel, an interactive R-Shiny package for automating predictive modeling and improving ML interpretability.	
AWARDS AND HONORS	• EvoSTAR 2025 Best Paper Award / Outstanding Student Award 2025 • PhD Studentship - UK Research and Innovation (UKRI) 2023 • 2nd place in Epsilon India Recommendation Engine Hackathon (26 Teams) 2020 • Top 6% Kaggle Categorical Feature Encoding Challenge (1,342 Teams) 2019 • Champion Innovator Spot Award, Epsilon 2019 • Top 3% Kaggle TalkingData Ad-Tracking Fraud Detection Challenge (3,946 Teams) 2018	
	GRANTS AWARDED "Using deep learning to build a robust automated recognition system for rare and endangered birds in India" <i>AI for Earth Microsoft Azure Research Credits Grant. \$5,000.</i> Oct 2021 • PI: Keshav Bhandari • Role: I substantially wrote and edited this grant, and it was based on my work.	
TEACHING	• Guest Lecturer & Demonstrator, Computational Creativity Jan 2025 – May 2025 5 Lecture Topics: Generative AI for Creative Language, Generative Music and Audio, Evolutionary Computation, and Evaluating Generative Systems.	
	• Senior Demonstrator, Data Science and AI master's cohort	Sep 2024 - April 2025
	• Teaching Fellow, Supervised 6 UG and 12 PG students on FYP	Oct 2024 - July 2025
	• Demonstrator, Computational Creativity	Jan 2025 - May 2025
	• Demonstrator, Statistics for AI and Data Science	Sep 2024 - Dec 2024
	• PhD Mentor, Google DeepMind Research Ready Programme	Jun 2024 - Jul 2024

INVITED TALKS
&
PRESENTATIONS

- **Invited Talk:** *ImprovNet - Generating Controllable Musical Improvisations with Iterative Corruption Refinement*. Northwestern University, McCormick School of Engineering. Feb. 13th, 2025.
- **Guest Lecture:** *Limitations of Sequential Models and An Introduction To Transformers*. Queen Mary University of London, Google Deepmind Research Ready Programme Orientation. June 26th, 2025.
- **Guest Lecture:** *Self-Supervised Corruption-Refinement Models for Low Resource Generative Music*. Queen Mary University of London, Class: Computational Creativity. April 2nd, 2025.
- **Guest Lecture:** *How To Be An Expert At Literature Reviews*. Queen Mary University of London, Google Deepmind Research Ready Programme Orientation. June 28th, 2024.
- **Research Presentation:** *Towards Melodic Development with Discrete Diffusion Models for Symbolic Music*. Queen Mary University of London, Digital Music Research Network One-Day Workshop (DMRN+18). December 2023.
- **Research Presentation:** *Augmentations to improve rare bird call classification for a highly imbalanced multi-label soundscape environment*. Digital University Kerala, 11th International Conference on Ecological Informatics (ICEI 2020+1). November 2021.

ACADEMIC &
COMMUNITY
SERVICE

Workshop Chair | Lead Organizer, 1st International Workshop on Emerging AI Technologies for Music (EAIM-26), AAAI-26 2026

Session Chair, Special Track: Human-AI Interaction in Creative Arts and Sciences, IJCNN 2025

Reviewer, IETE Technical Review 2026

Reviewer, The Association for the Advancement of Artificial Intelligence (AAAI) 2025

Reviewer, International Joint Conference on Neural Networks (IJCNN) 2025

Member, Data Kind, Bangalore Chapter Aug 2018-Oct 2019

President, Purdue Marketing Association, Purdue University Jan 2015-Dec 2015